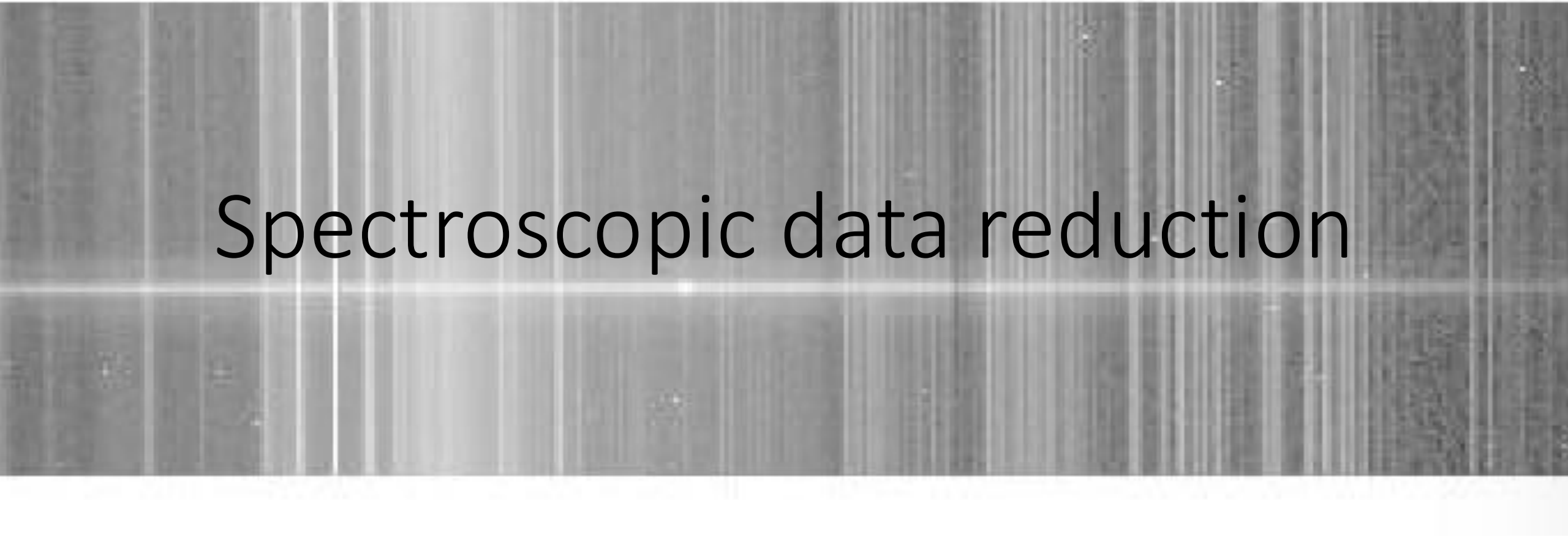
ACME
Astrophysics Centre for Multimessenger studies in Europe



Spectroscopic data reduction

Irene Salmaso

Diffraction from single slit

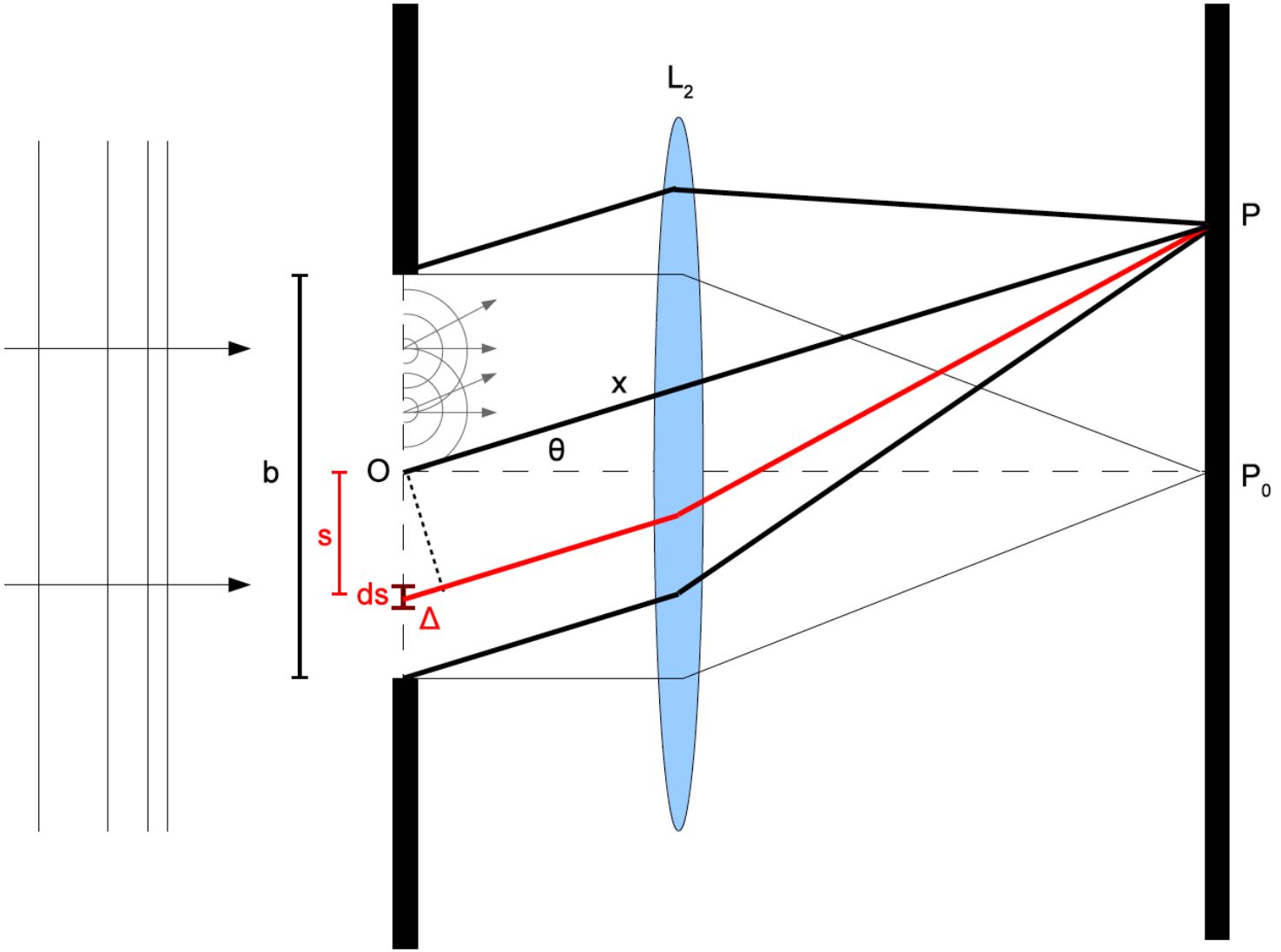


Image credits: Stefano Ciroi

Diffraction figure

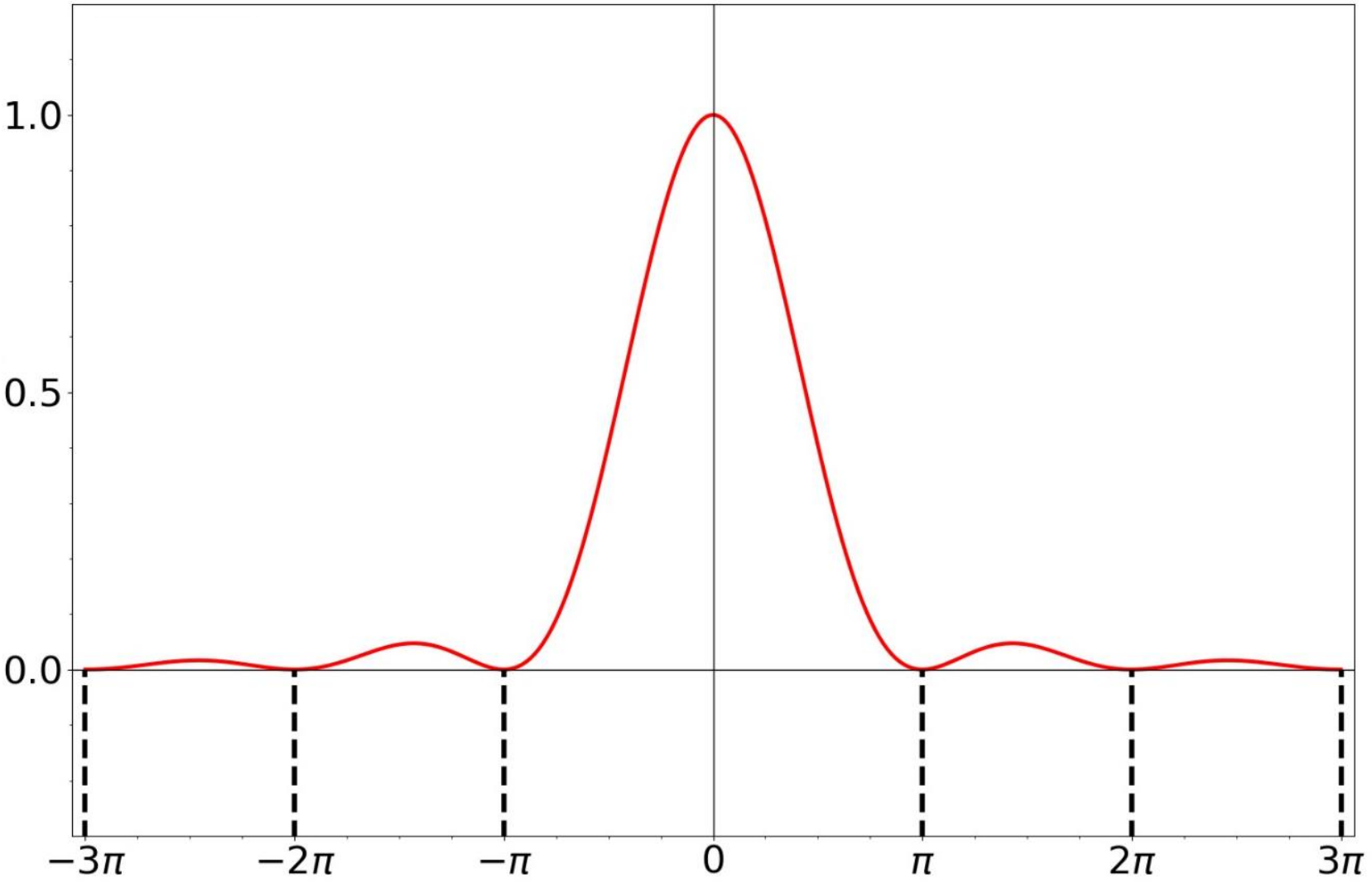


Image credits: Silvia Conforti

Diffraction from two slits

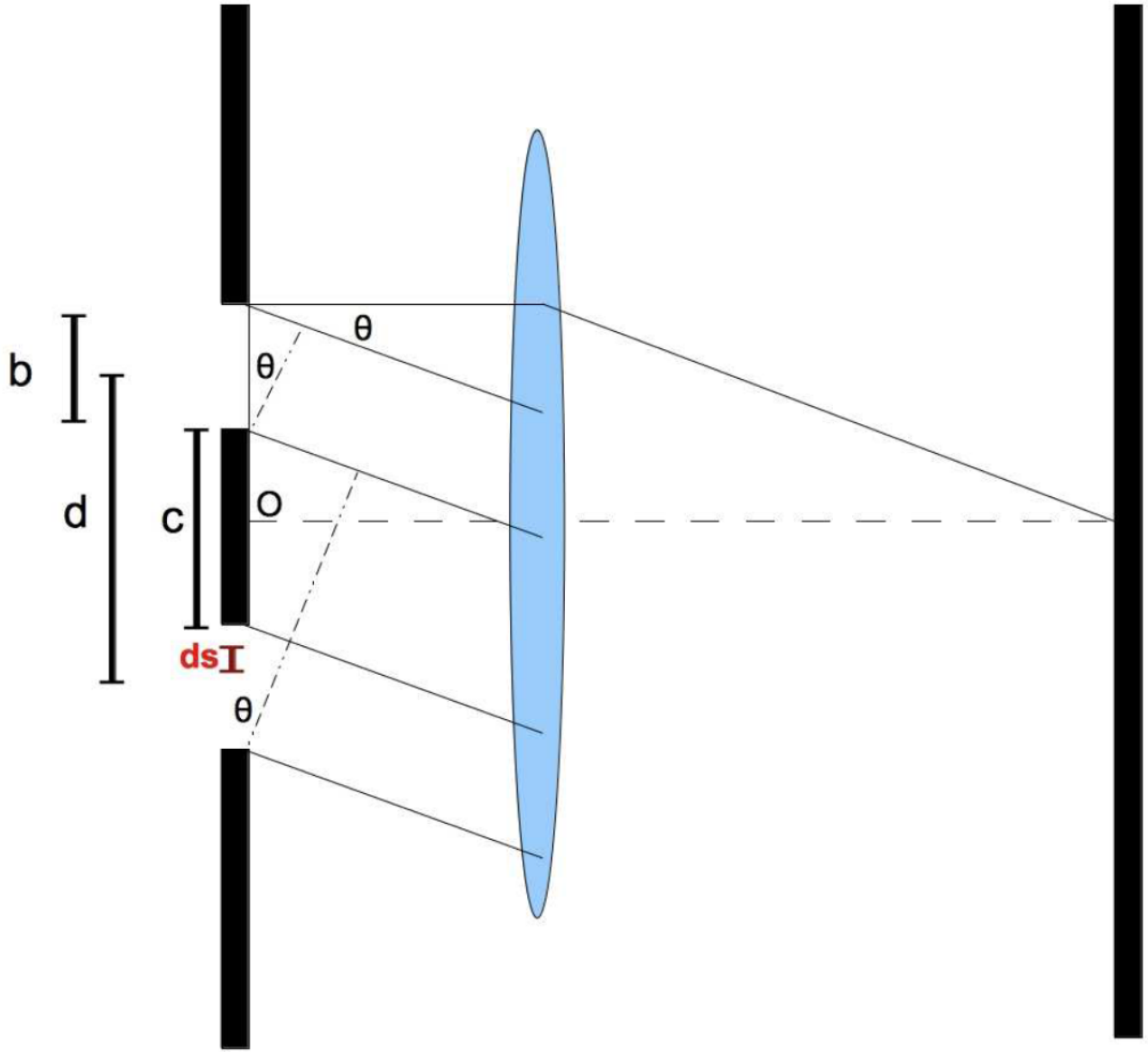


Image credits: Stefano Ciroi

Diffraction figure

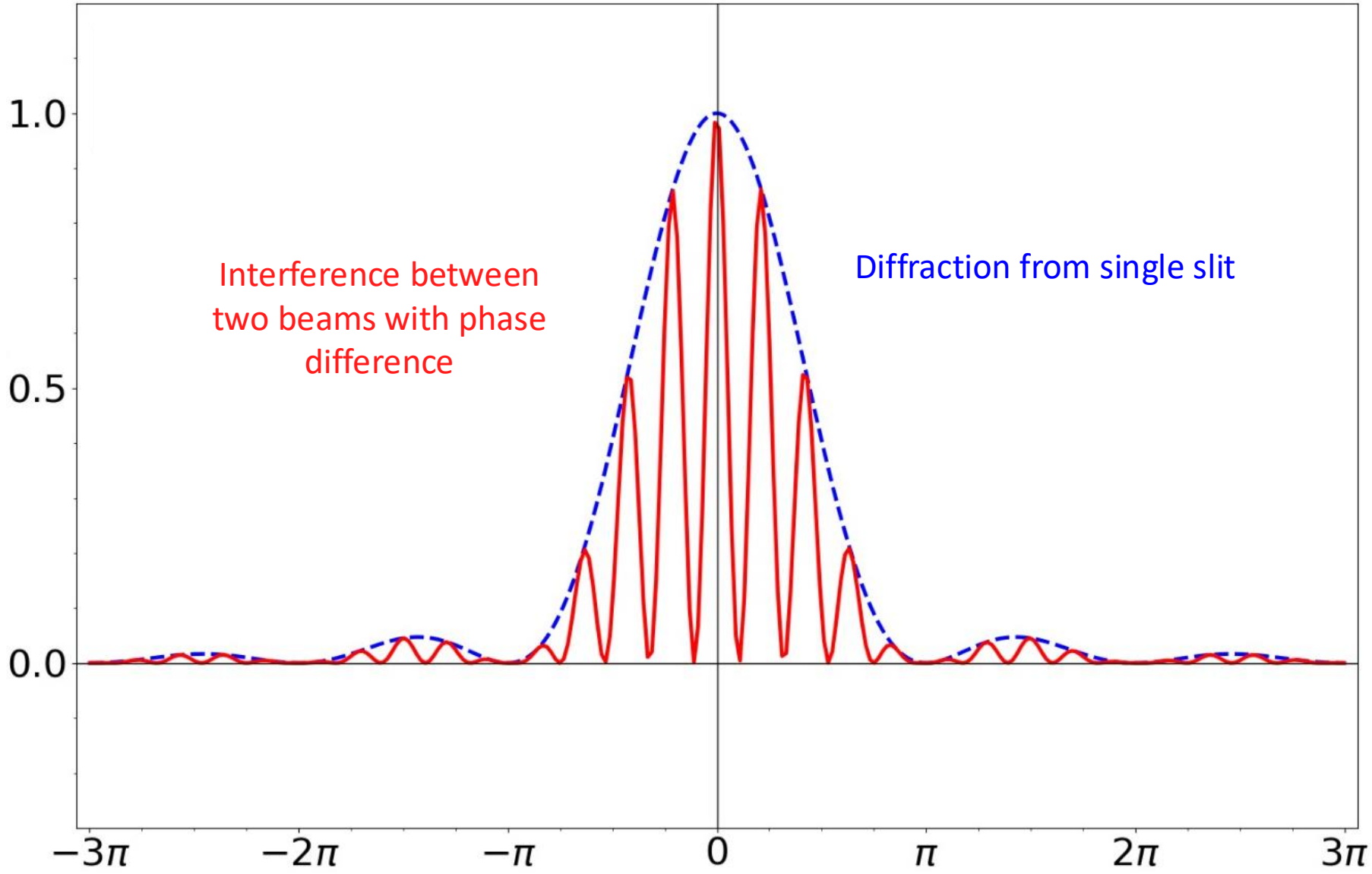


Image credits: Silvia Conforti

Diffraction from N slits

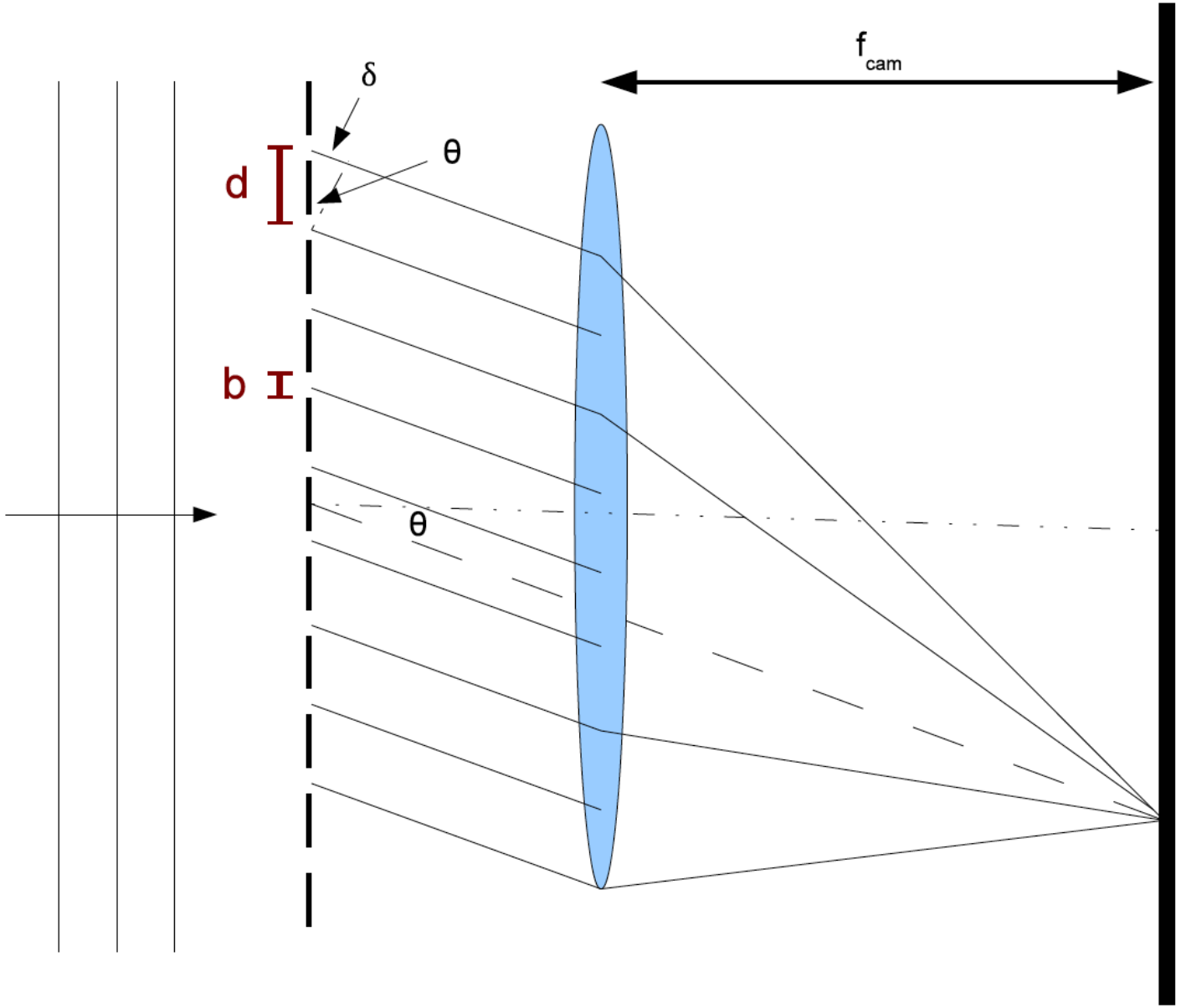


Image credits: Stefano Ciroi

Main maxima = spectral lines

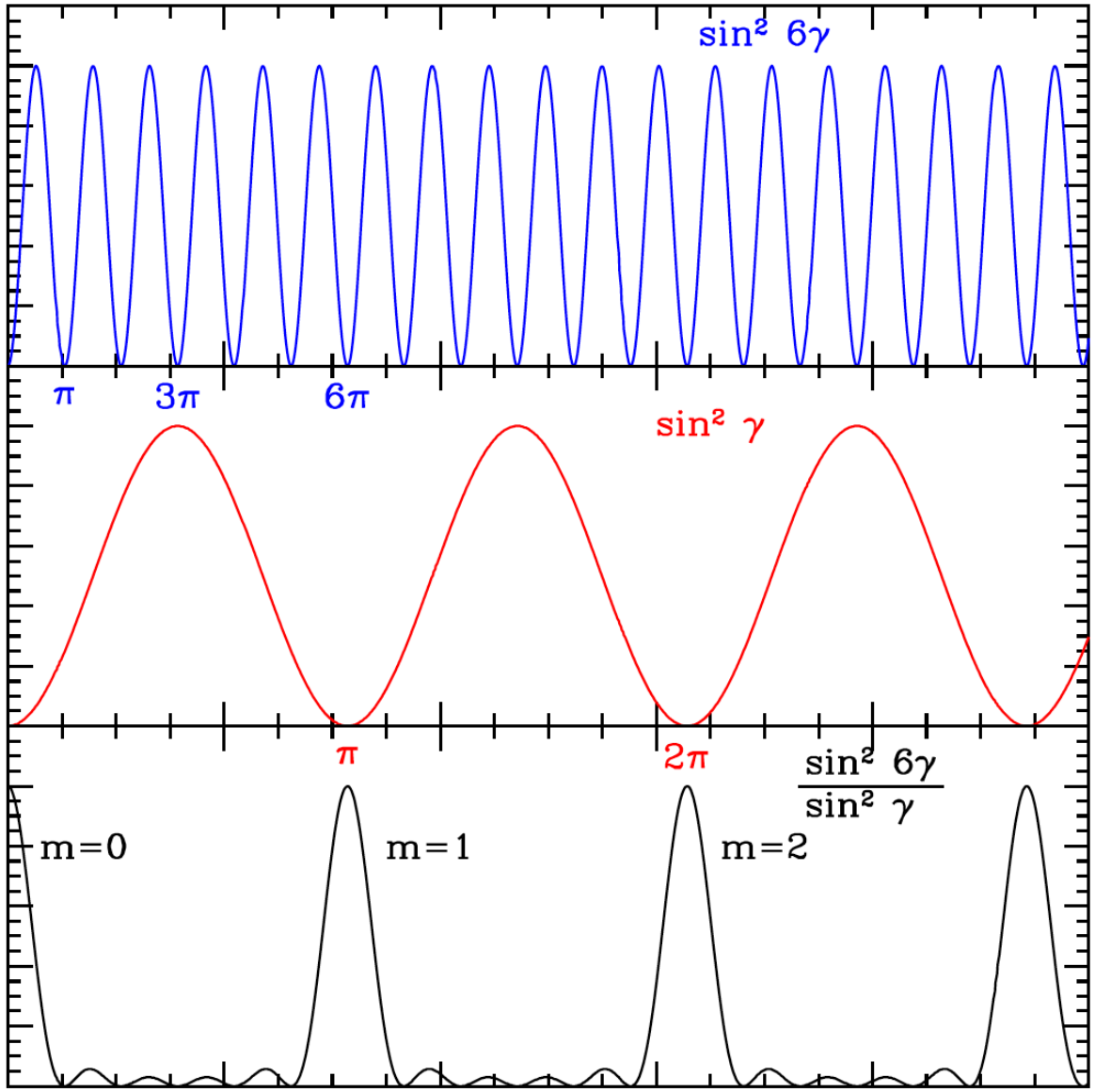
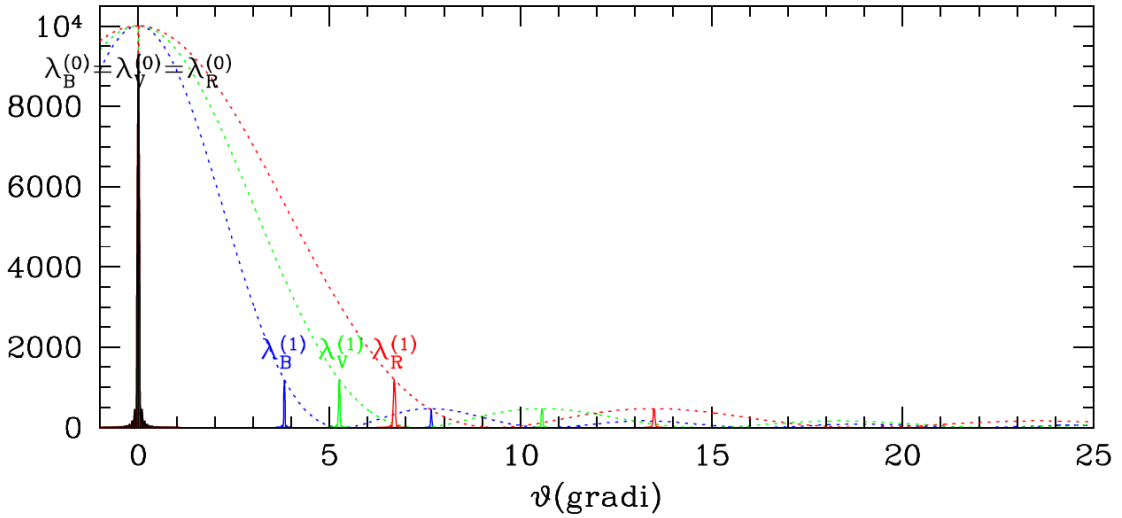


Image credits: Stefano Ciroi

Dispersion (formation of the spectrum)



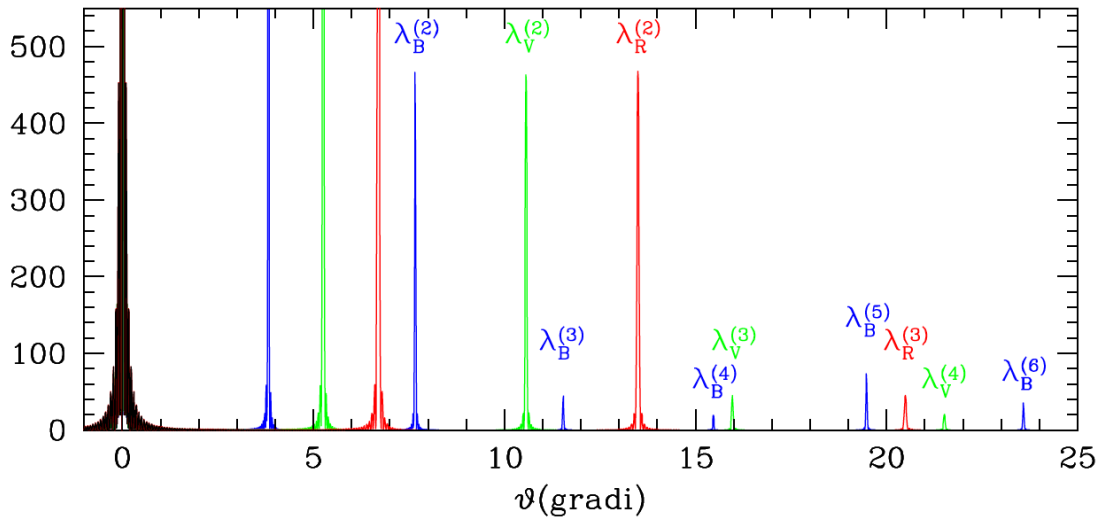
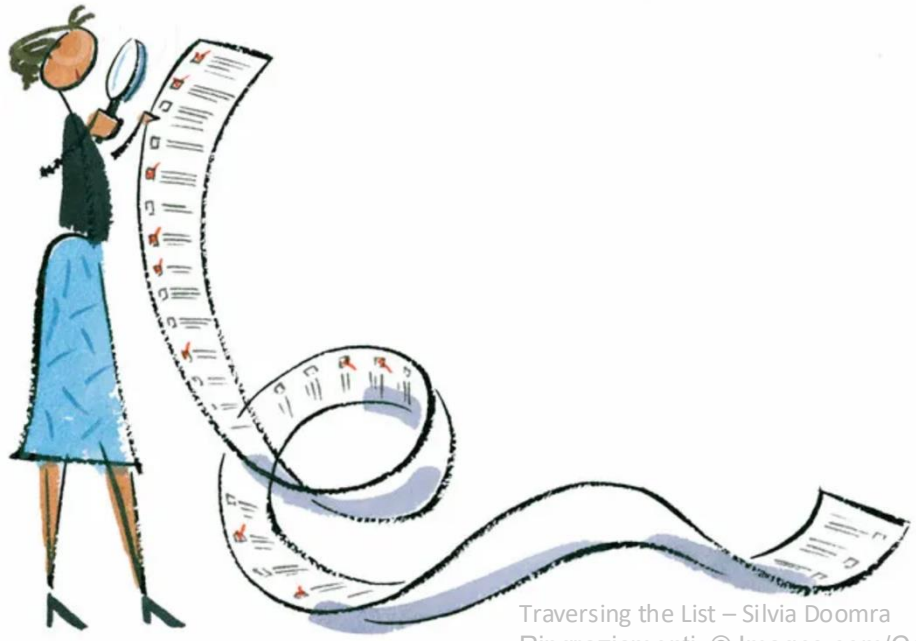


Image credits: Stefano Ciroi

(standard) Reduction of spectroscopi data

1. Bias subtraction
2. Flat-field correction
3. Cosmic-ray rejection (optional)
4. Wavelength calibration
5. Flux calibration
6. Atmospheric correction



Traversing the List – Silvia Doomra
Ringraziamenti: © Images.com/Corbis
Copyright: © Corbis. All Rights Reserved.

IRAF (Image Reduction and Analysis Facility)

```
This package contains tasks for viewing and manipulating 3d images.
It is a pre-release version, and does not reflect the ultimate
partitioning of n-dimensional image tasks within IRAF

  NOAO/IRAF PC-IRAF Revision 2.16.1+ DEBIAN
      This is the DEBIAN version of IRAF V2.16+ supporting PC systems.

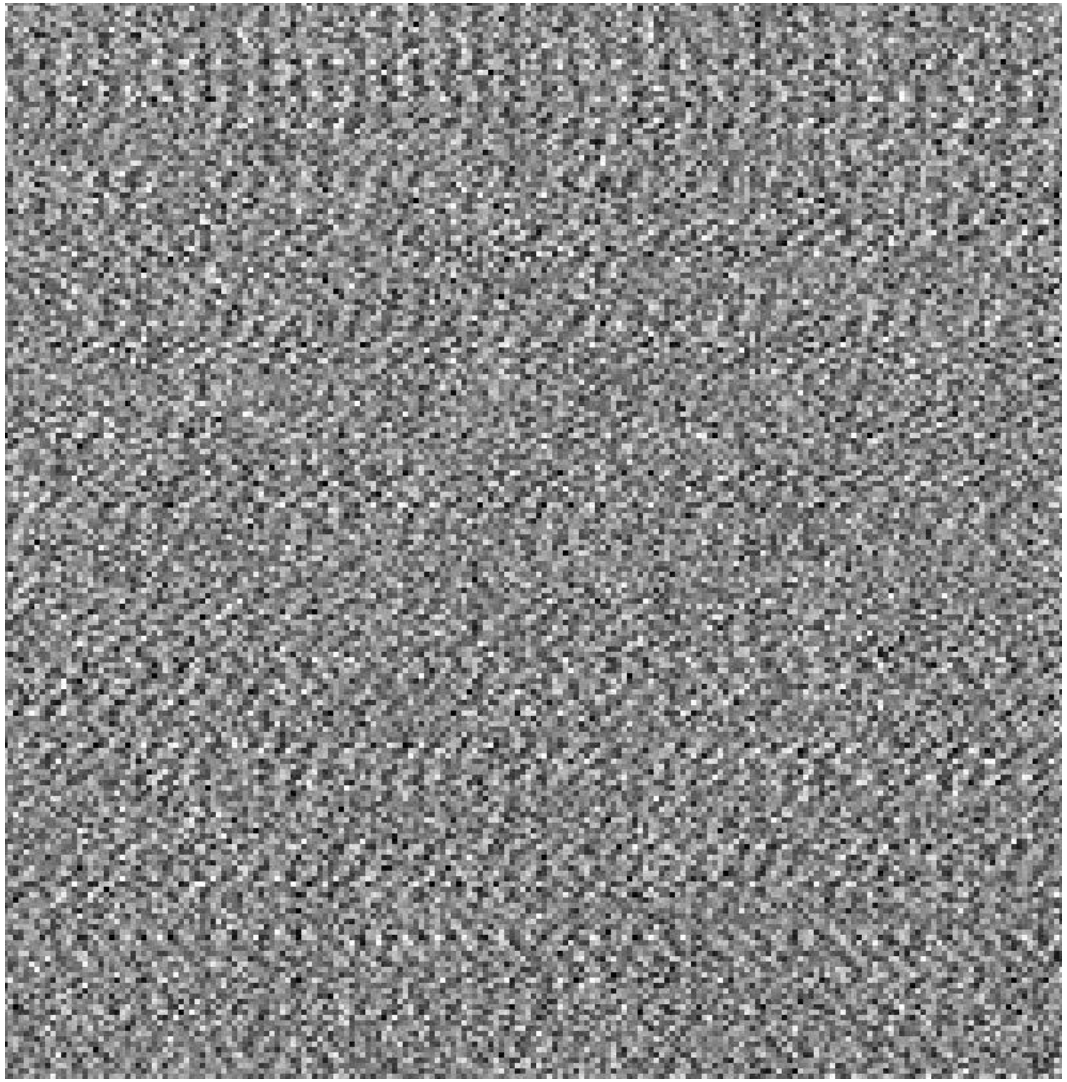
  This product includes results achieved by the IRAF64 project in 2006-
  2009 directed by Chisato Yamauchi (C-SODA/ISAS/JAXA).

  Welcome to IRAF.  To list the available commands, type ? or ??.  To get
  detailed information about a command, type `help <command>'.  To run  a
  command  or  load  a  package,  type  its name.   Type  `bye' to exit a
  package, or `logout' to get out  of the CL.    Type `news' to find  out
  what is new in the version of the system you are using.

  Visit http://iraf.net if you have questions or to report problems.

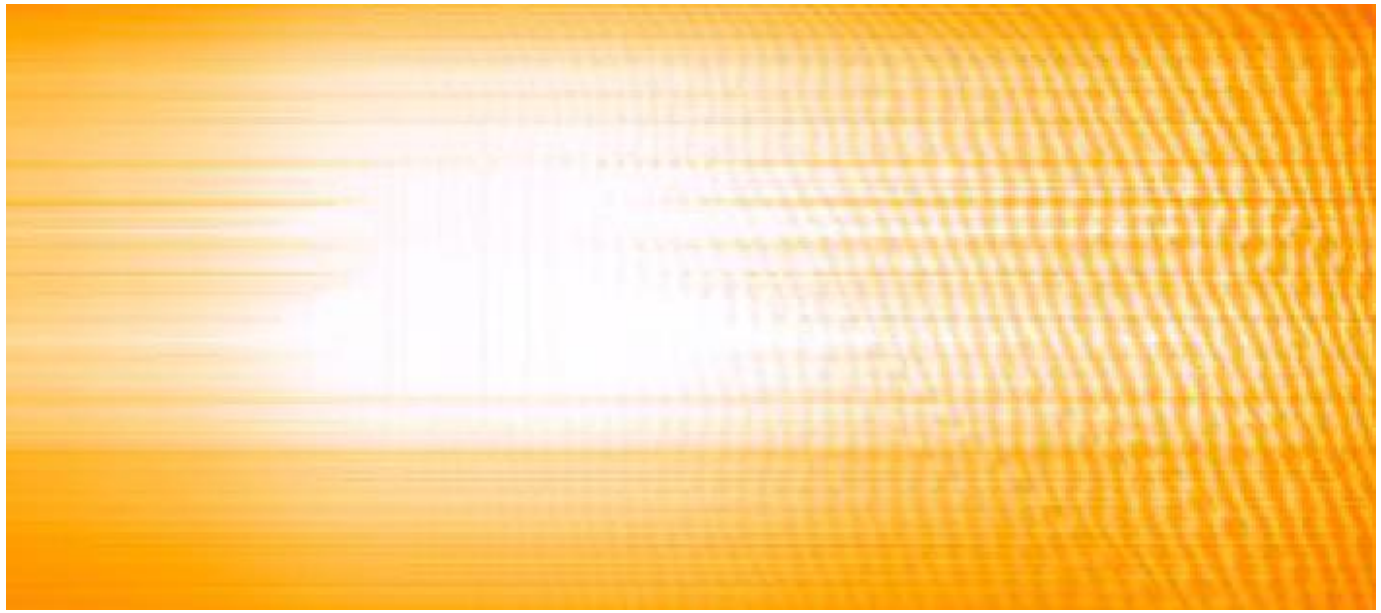
  A common global login file is  available at /etc/iraf/login.cl. You may
  consider removing or replacing your personal login script.
clpackage/:
 clpackage/     language/       plot/           system/
 dataio/        lists/          proto/          user/
 dbms/          noao/           rvsao/          utilities/
```

1. Bias subtraction



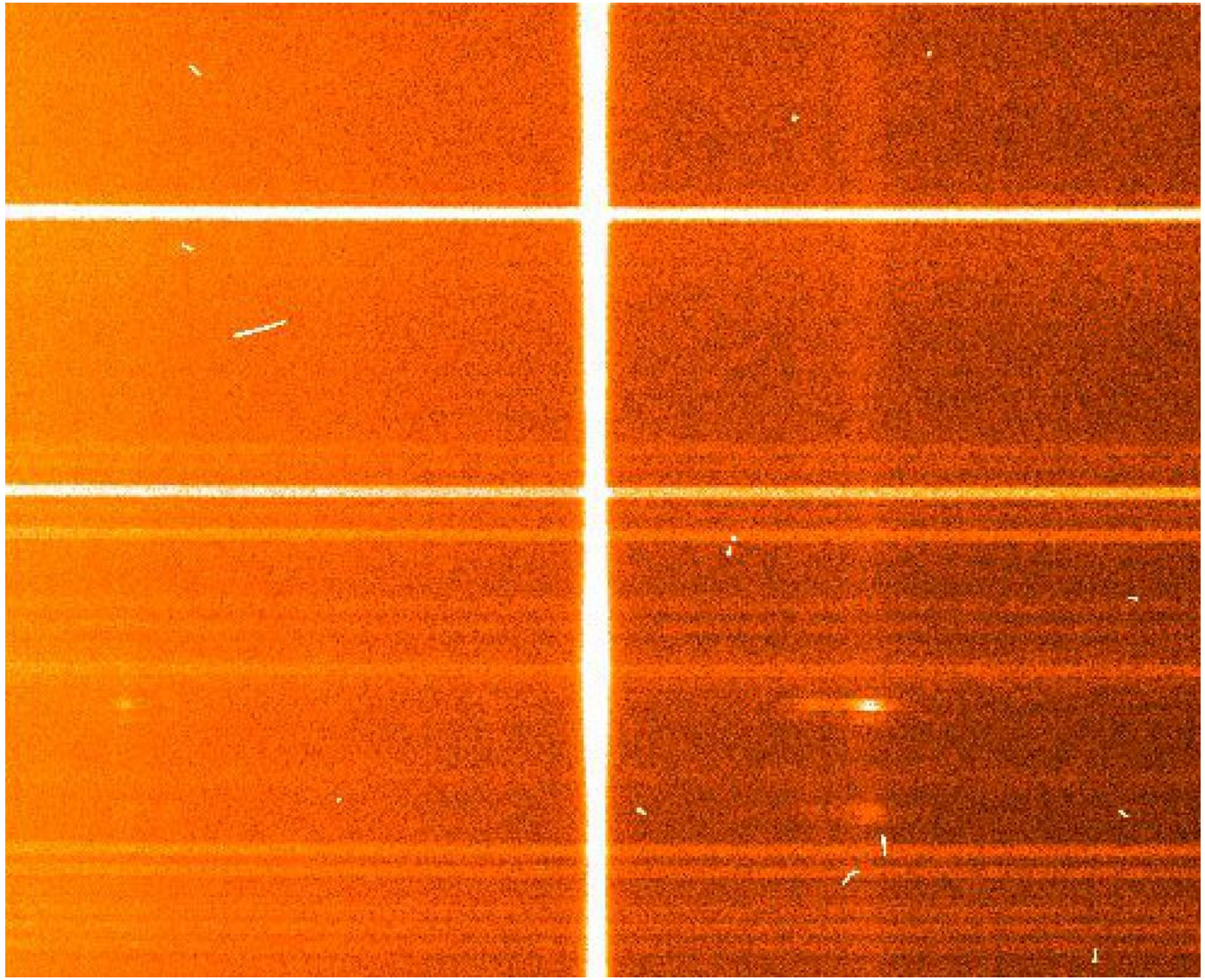
- `ccdred` trimming
- `zerocombine` combine
- `ccdproc` apply to flats

2. Flat-field correction

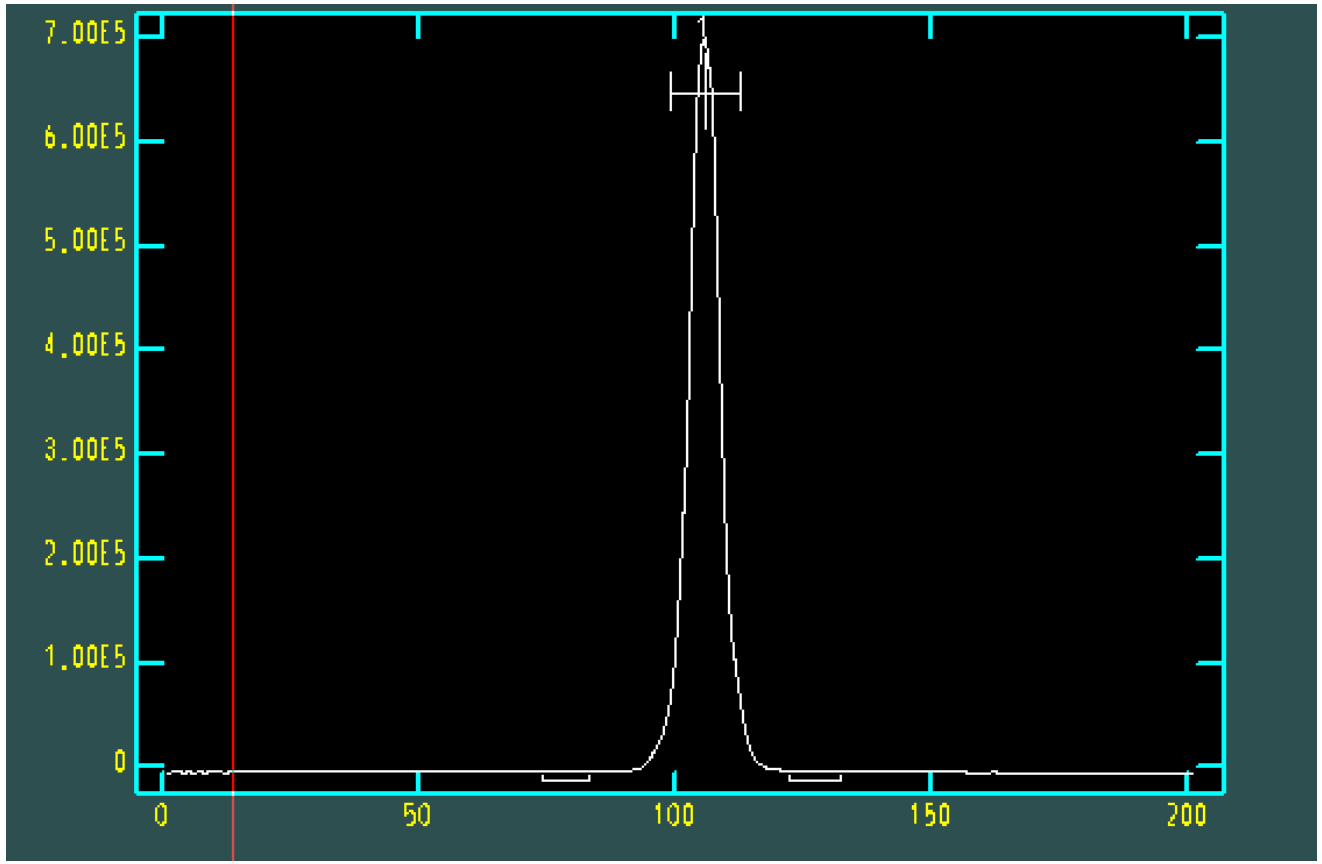


- `ccdproc` bias subtraction
- `flatcombine` combine
- `response` normalise to 1
- `ccdproc` apply to arcs and science

3. Cosmic-ray rejection

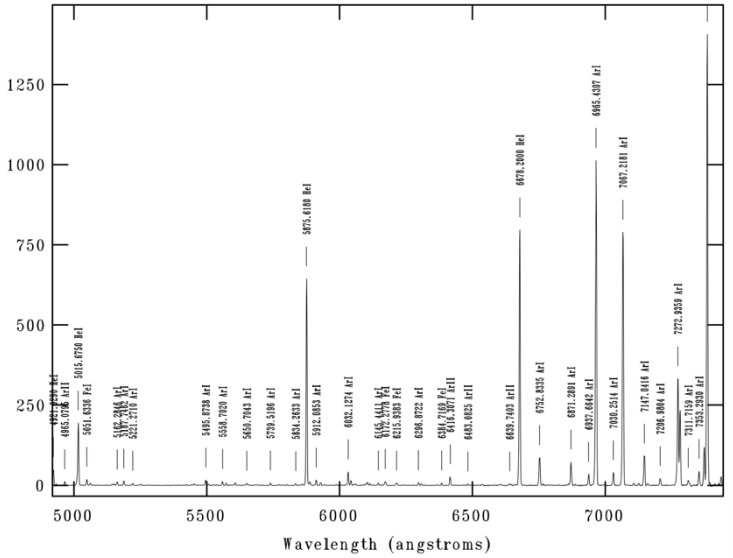


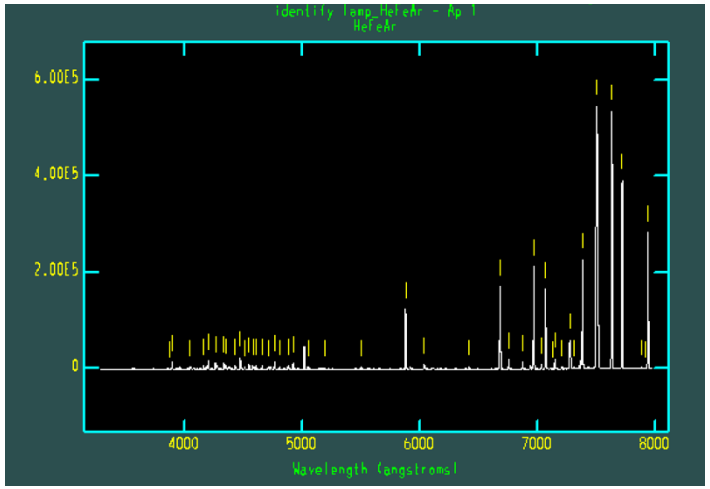
3.1 Extract the spectrum



- `apall` extract spectra of science images

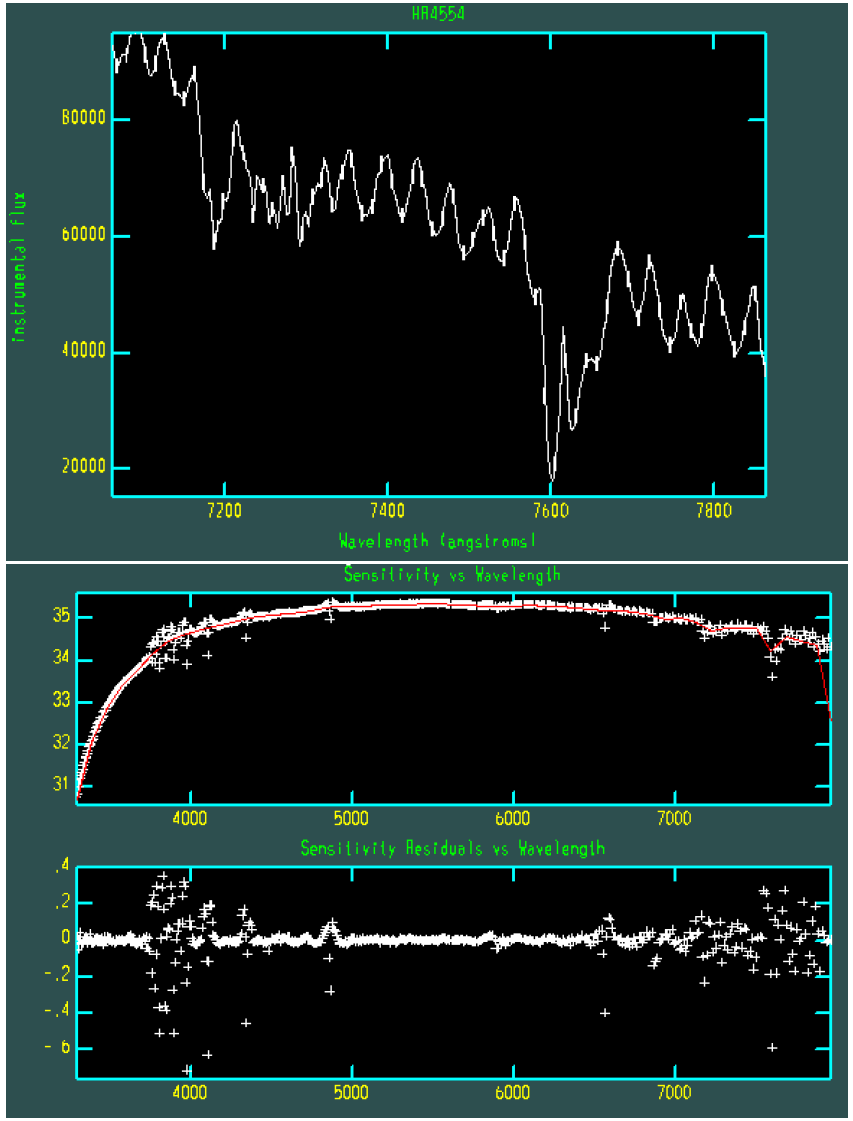
4. Wavelength calibration





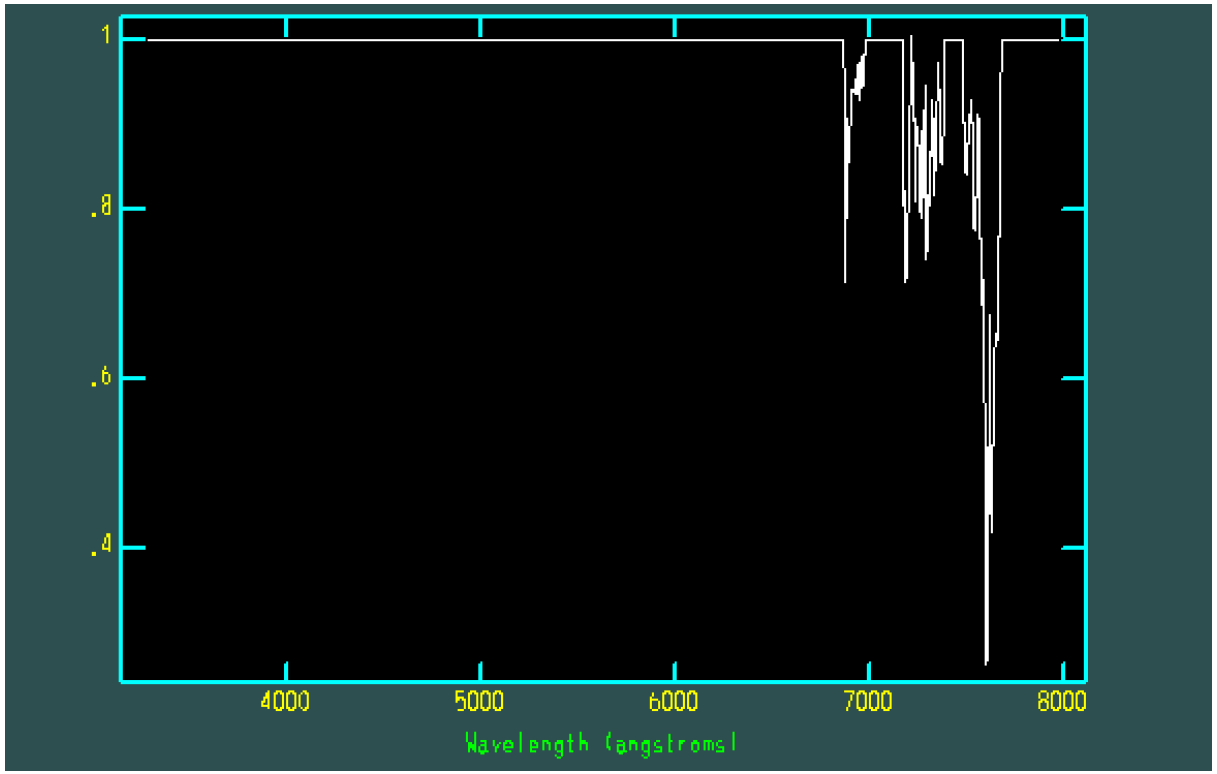
- `apsum` 1D spectra of arcs (in `reference` object spectrum before apall)
- `identify` connect pixel and wavelength
- `refspec` associate identification to objects (in `reference` 1D arc)
- `dispcor` associate to objects for good

5. Flux calibration



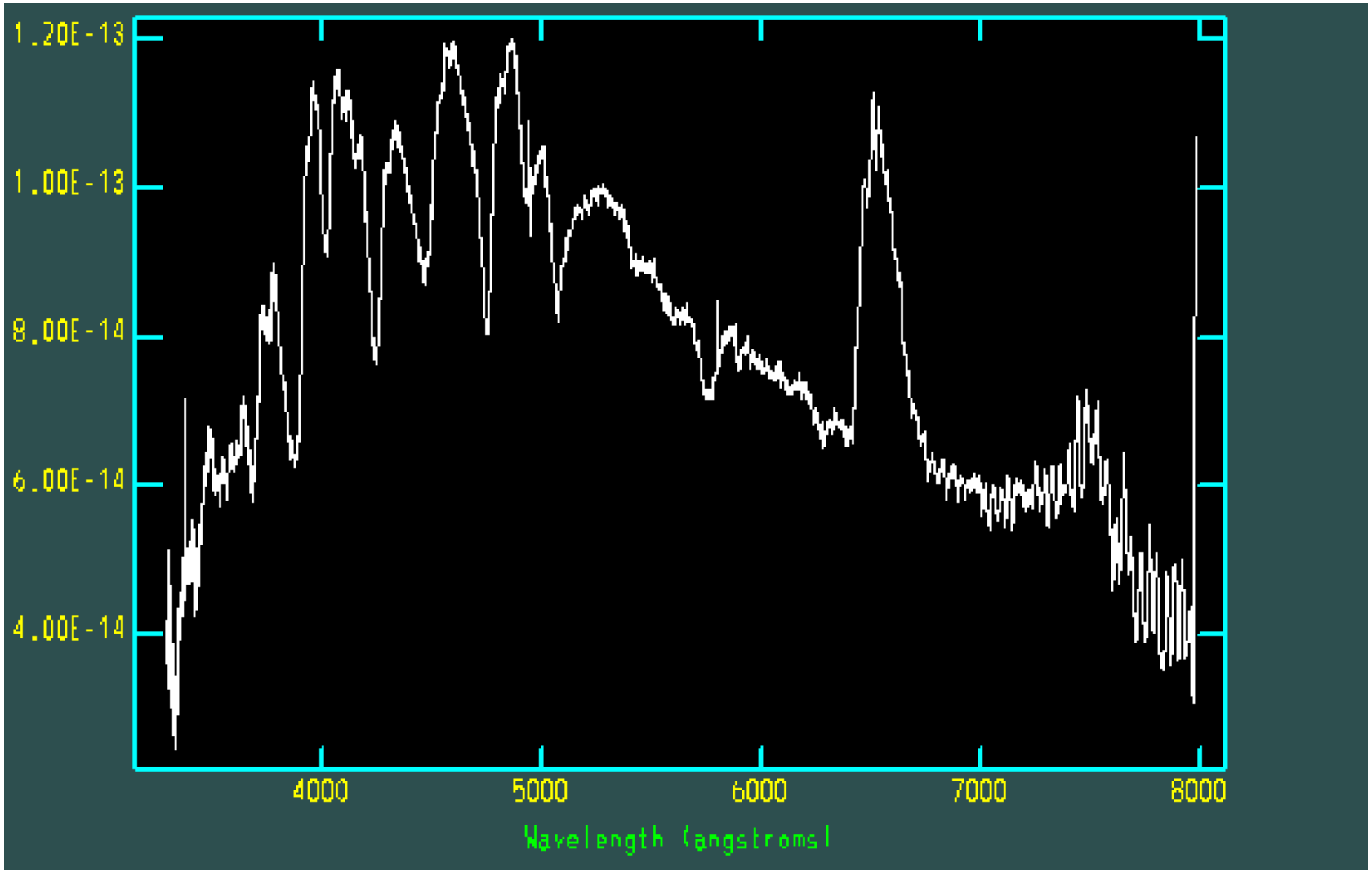
- `standard` compare standard star spectrum to a flux table
- `sensfunc` compute calibration curve
- `calib` calibrate the object

6. Atmospheric correction

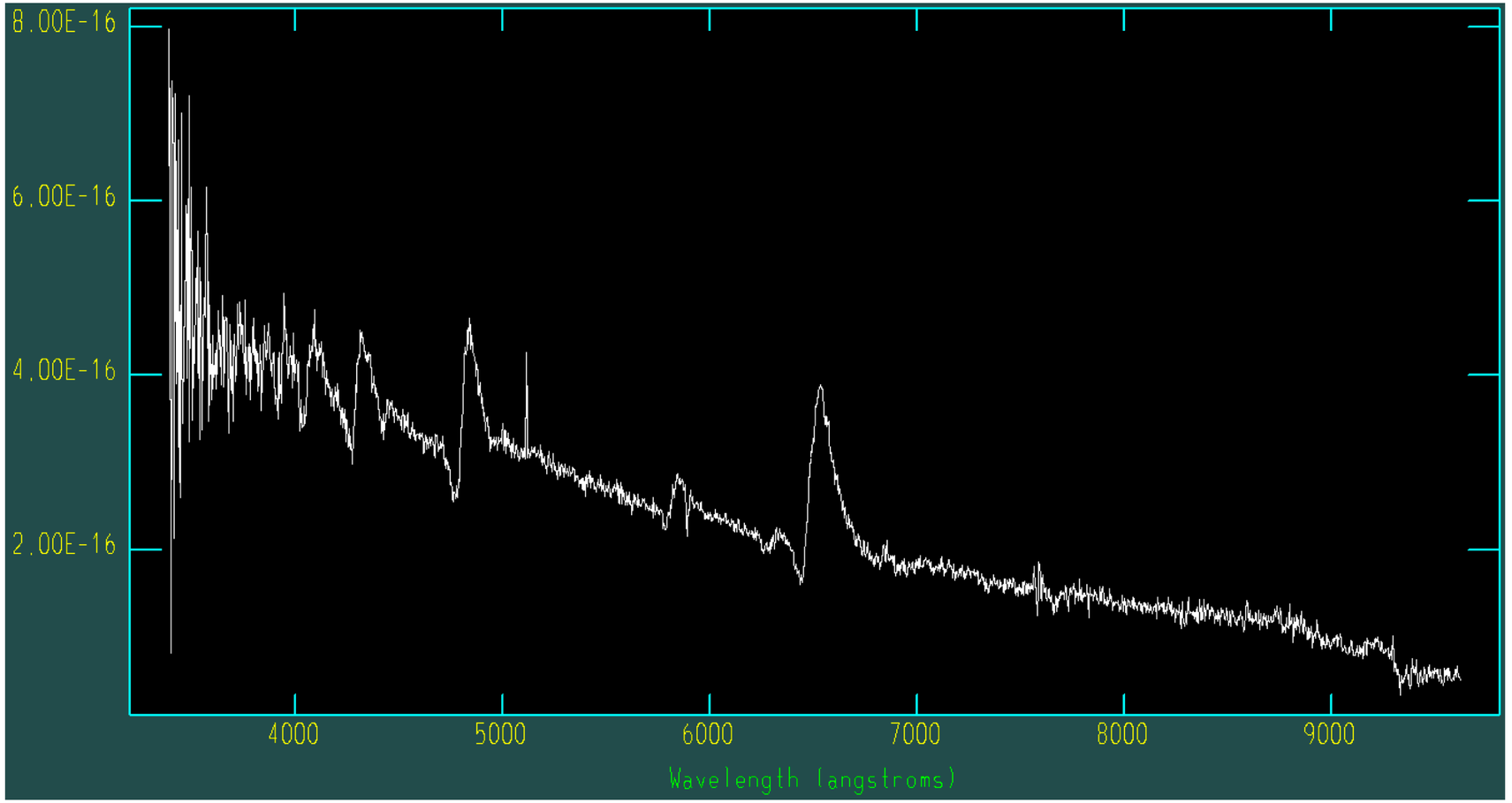


- `splot` plot the extracted spectrum and zap telluric lines
- `telluric` apply atmospheric correction to the object spectrum

Enjoy your spectrum!



We got a spectrum! Now what?



>> splot your_file.fits

Cursor location
is important!!

Zoom:	W + E + E
De-zoom:	W + A
EW:	E + E
Gaussian fit:	K + G
Deblend:	D + D
Quit:	Q